LEVEL-VI

SINGLE ANSWER QUESTIONS

1. Let $f: R \to R$ defined by $f(x)=\frac{e^{x^2}-e^{-x^2}}{e^{x^2}+e^{-x^2}}$ then f(x) is
a) one-one but not onto
b)neither one-one nor onto
c)many-one but onto d)one-one but not onto

2. If the graph of a function f(x) is symmetrical about the line x=a then
a) $f(a+x)=f(a-x)$
b) $f(a+x)=f(x-a)$
c) $f(x)=f(-x)$ d) $f(x)=-f(-x)$

3. If $f(x)=f(2a-x)$ then the graph of f(x) is symmetric about the line
a)y=a b)x=2a c)x=a d)x=-a

4. There are exactly two linear functions which map from [-1,1] onto [0,2] they are
a) $y=x+1, y=x-1$ b) $y=x+1, y=-x+1$
c) $y=2x-1,\ y=x-4$ d) $y=x,\ y=3x$

5. $f:(-\infty,-1] \to (0,e^5]$ defined by $f(x)=e^{x^3-3x+2}$ is
a)one-one and into b)one-one and onto
c)many-one and into d)many-one and onto

6. Let $f(x)=x+2|x+1|+2|x-1|$. If $f(x)=k$ has exactly one real solution, then k equals
a) 3 b) 0 c) 1 d) 2

7. Let $f(1)=1$ and $f(n)=2\sum_{r=1}^{n-1} f(r)$ then $\sum_{n=1}^{m} f(n)=$
a) 3^m-1 b) 3^m c) 3^{m-1} d) 3^{m-2}

8. The domain of $f(x)=\log_{10}\left\{1-\log_{10}\left(x^2-5x+16\right)\right\}$ is
a) (2,3) b) $(0,\infty)$ c) $[1,3]$ d) $[2,3]$

9. If [.] denotes G.I.F then the domain of $f(x)=\cos^{-1}\left(x+[x]\right)$ is
a) (0,1) b) [0,1) c)[0,1] d)[-1,1]

10. The domain of $f(x)=\cot^{-1}\left(\frac{x}{\sqrt{x^2-[x^2]}}\right)$ (where [.] is G.I.F)
a)R b) $R-\{0\}$
c) $R-\left\{\pm\sqrt{n},\ n\in Z \ and \ n\geq 0\right\}$
d)None

11. The domain of $f(x)=\frac{1}{\ln\left[\cos^{-1}x\right]}$ (where [.] G.I.F) is
a)[0,1] b) $[-1,\cos 2]$
c) $[-1,\cos 3)\cup(\cos 3,\cos 4)$
d) $[-1,\cos 3)\cup(\cos 3,\cos 2)$

12. If $f(x)=\cos^{-1}\left(x-x^2\right)+\sqrt{\left(1-\frac{1}{|x|}\right)}+\frac{1}{\left[x^2-1\right]}$ then domain of $f(x)$ (where [.] G.I.F) is
a) $\left(\sqrt{2},\frac{1+\sqrt{5}}{2}\right]$ b) $\left(-\sqrt{2},\frac{1-\sqrt{5}}{2}\right)$
c) $\left[\sqrt{2},\frac{1+\sqrt{5}}{2}\right]$ d) $\left[\sqrt{7},\frac{\sqrt{7}}{2}\right]$

13. The range of $f(x)=\sin^{-1}\left(\sqrt{x^2+x+1}\right)$ is
a) $\left[0,\frac{\pi}{2}\right]$ b) $\left[0,\frac{\pi}{3}\right]$ c) $\left[\frac{\pi}{3},\frac{\pi}{2}\right]$ d) $\left[\frac{\pi}{6},\frac{\pi}{2}\right]$

14. The function $f(x)=|px-q|+r|x|, x\in(-\infty,\infty)$ where p>0,q>0,r>0 assumes its minimum value only at one point if
a) $p\neq q$ b) $r\neq q$ c) $r\neq p$ d) $p=q=r$

15. Let f,g,h be real -valued functions defined on the interval [0,1] by $f(x)=e^{x^2}+e^{-x^2}$; $g(x)=x.e^{x^2}+e^{-x^2}$ and $h(x)=x^2.e^{x^2}+e^{-x^2}$. If a,b,c denote respectively the absolute max.values of f,g,h on [0,1] then

a) $a=b$ and $c\neq b$ b) $a=c$ and $a\neq b$
c) $a\neq b$ and $c\neq b$ d) a=b=c

16. The range of $f(x)=\cos^{-1}\left(\frac{\sqrt{2x^2+1}}{x^2+1}\right)$ is

a) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ b) $[0,\pi]$ c) $[0,\frac{\pi}{2})$ d) $(0,\pi)$

17. Let 'f' be an injective mapping with domain $\{x,y,z\}$ and range $\{1,2,3\}$ such that exactly one of the following statements is correct and the remaining are false $f(x)=1; f(y)\neq 1; f(z)\neq 2$ then $f^{-1}(1)=$ [IIT 1982]

a) x b) y c) z d) None

18. If $f(x)=-1+|x-2|; 0\leq x\leq 4$
$g(x)=2-|x|; -1\leq x\leq 3$ then $(fog)(x)=$

a) $\begin{matrix}-1-x; -1\leq x\leq 0\\ x-1\ ;\ 0<x\leq 2\end{matrix}$ b) $\begin{matrix}-1+x; 0\leq x\leq 1\\ x-1\ ;\ 1<x\leq 2\end{matrix}$

c) $-1-x; -1\leq x\leq 2$ d) does not exist

19. If $f:[1,\infty)\to[2,\infty)$ is given by $f(x)=x+\frac{1}{x}$ then $f^{-1}(x)=$

a) $\frac{x\pm\sqrt{x^2-3}}{4}$ b) $\frac{x\pm\sqrt{x^2-4}}{2}$

c) $\frac{x+\sqrt{x^2-4}}{2}$ d) $\frac{x-\sqrt{x^2-4}}{2}$

20. The period of the real - valued function satisfying $f(x)+f(x+4)=f(x+2)+f(x+6)$ is

a) 10 b) 8 c) 12 d) 6

21. Let f(x) be a periodic function with period 3 and $f\left(\frac{-2}{3}\right)=7$ and $g(x)=\int_0^x f(t+n)dt$ where $n=3K, K\in N$ then $g^1\left(\frac{7}{3}\right)=$

a) $\frac{-2}{3}$ b) 7 c) -7 d) $\frac{7}{3}$

22. If $f(x)$ is an even function and satisfies the relation $x^2.f(x)-2.f\left(\frac{1}{x}\right)=g(x)$ where $g(x)$ is an odd function, then the value of $f(5)$ is

a) 0 b) $\frac{37}{55}$ c) 4 d) 5

23. Consider a real valued function f(x) satisfying $2f(xy)=(f(x))^y+(f(y))^x$ $\forall x,y\in R$ and $f(1)=a$ where $a\neq 1$ then $(a-1)\sum_{i=1}^{n}f(i)=$

a) $a^{n+1}+a$ b) a^{n+1} c) $a^{n+1}-a$ d) $a^{n+2}-a$

24. If $'p'$ and $'q'$ are +ve integers, f is a function defined for +ve numbers and attains only +ve values such that $f(x,f(y))=x^p y^q$ then $p^2=$

a) 2q b) q c) 3q d) 4q

25. If the function 'f' satisfies the relation $f(x+y)+f(x-y)$ $=2f(x)f(y)\ \forall x,y\in R$ and $f(0)\neq 0$ then f(x) is an

a) even function
b) odd function
c) Neither even nor odd
d) can not decide

26. A real valued function f(x) satisfies the function $f(x-y)=f(x).f(y)-f(a-x)f(a+y)$ where 'a' is a given constant and $f(0)=1$,

then the graph of the function is symmetrical about [IIT 2005]

a) $point(2a,0)$ b) $point(a,0)$

c) $line\ x=2a$ d) $line\ x=a$

27. **The function 'f' satisfies the functional equation** $3f(x)+2f\left(\frac{x+59}{x-1}\right)=10x+30$ **for all real** $x\neq 1$**, then the value of** $f(7)$ **is**

a) 8 b) 4 c) -8 d) 11

28. **Let 'f' be a real valued function defined for all** $x\in R$ **such that for some fixed a>0,** $f(x+a)=\frac{1}{2}+\sqrt{f(x)-(f(x))^2}$ **for all** 'x' **then the period of** $f(x)$ **is**

a) $\frac{a}{4}$ b) $\frac{a}{3}$ c) $2a$ d) None

29. **Let** $f(x,y)$ **be a periodic function satisfying** $f(x,y)=f(2x+2y,2y-2x)\forall x,y\in R$ **defined** $g(x)=f(2^x,0)$ **then the period of** $g(x)$ **is**

a) 4 b) 6 c) 8 d) 12

30. **If** $f(x)=\min\left(x^2,x,\ \mathrm{sgn}\left(x^2+4x+5\right)\right)$ **then the value of** $f(2)$ **is equal to**

a) 2 b) 0 c) 1 d) 4

31. **If** $f(x)=\sin^2 x+\sin^2\left(\frac{\pi}{3}+x\right)+\cos(x)\cos\left(x+\frac{\pi}{3}\right)$ **and** $g\left(\frac{5}{4}\right)=1$ **then the graph of** $y=g(f(x))$ **is**

a) a circle b) a straight line

c) a parabola d) a pair of straight lines

32. **Let** $f_1(n)=1+\frac{1}{2}+\frac{1}{3}+-----+\frac{1}{n}$ **then** $f_1(1)+f_1(2)+f_1(3)+------+f_1(n)=$

a) $n.f_1(n)-1$ b) $(n+1)f_1(n)+n$

c) $(n+1)f_1(n)-n$ d) $n.f_1(n)+n$

MULTIPLE ANSWER QUESTIONS

33. **If** $f(x)=x|x|;\ \ 0\le x<1$
$=2x\ ;\ x\ge 1$ **then its**

a) even extension is $x^2;-1\le x\le 0$, $-2x;-\infty<x\le -1$

b) odd extension is $x^2;-1\le x\le 0$, $2x;-\infty<x\le -1$

c) even extension $x^2;0\le x\le 1$, $-2x;1<x<\infty$

d) does not exist

34. **If** $f(x)=\sqrt{3|x|-x-2}$ **and** $g(x)=\sin x$ **then the domain of** $(fog)(x)=$

a) $\left\{2m\pi+\frac{\pi}{2}\right\};m\in z$

b) $\left[2m\pi+\frac{7\pi}{6},2m\pi+\frac{11\pi}{6}\right]\ m\in Z$

c) $\left\{2m\pi+\frac{\pi}{3}\right\},m\in z$

d) ϕ

35. **The domain of** $f(x)=\frac{1}{\sqrt{|[|x|-1]|-5}}$ **(where [.] G.I.F)is**

a) $[-7,7]$ b) $(-\infty,7]$

c) $(-\infty,-7]$ d) $[7,\infty)$

36. **For the function** $f(x)$ **satisfying** $2.f(\sin x)+f(\cos x)=x,\forall x\in R$

a) Domain is [0,1] b) range is $\left[\frac{-2\pi}{3},\frac{\pi}{3}\right]$

c) Domain is [-1,1] d) range is $\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$

37. **Let** $f(x)=\sin x, g(x)=\ln|x|$ **If the ranges of fog and gof are** R_1 *and* R_2 **respectively then** **[IIT 1994]**

a) $R_1=\{U:-1\le U\le 1\}$

b) $R_2=\{V:-\infty<V\le 0\}$

c) $R_1=\{U:0\le U\le 1\}$ d)None

38. **The polynomial p(x) is such that for any polynomial q(x) we have** $p(q(x))=q(p(x))$ **then** $p(x)$ **is**

a) even b)odd
c)of even degree d)of odd degree

39. **Let** $f(x)=\max\{1+\sin x,1,1-\cos x\}$; $x\in[0,2\pi]$ **and** $g(x)=\max\{1,|x-1|\}$; $x\in R$ **then**

a) $g(f(0))=1$ b) $g(f(1))=1$

c) $f(g(1))=1$ d) $f(g(0))=\sin 1$

40. **Let** $f(x)=\sin x+\cos\left(\left(\sqrt{4-a^2}\right)x\right)$ **. Then the integral values of 'a' for which f(x) is a periodic function , are given by**

a)1 b) 2 c) -2 d)0

41. **If a polynomial of degree 'n' satisfies** $f(x)=f^1(x).f^{11}(x)\forall n\in R$ **then f(x) is**

a) an onto function b) an into function
c) no such function is possible
d) even function

42. **Let** $f(x)=\ln(2x-x^2)+\sin\left(\frac{\pi x}{2}\right)$ **then**

a) graph of 'f' is symmetrical about the line x=1
b) graph of 'f' is symmetrical about the line x=2
c)max. value of 'f' is 1
d)min.value of 'f' does not exist

PARAGRAPH QUESTIONS

(P) **Let** $F(x)=f(x)+g(x)$;

$G(x)=f(x)-g(x)$ **and** $H(x)=\frac{f(x)}{g(x)}$

where $f(x)=1-2\sin^2 x$ **and** $g(x)=\cos(2x)\forall f:R\to[-1,1]$ **and** $g:R\to[-1,1]$

Now answer the following:

43. **Domain and range of** $H(x)$ **are**

a) R *and* $\{1\}$ b) R *and* $\{0,1\}$

c) $R-\left\{(2n+1)\frac{\pi}{4}\right\}$ and $\{1\}$ d)None

$n\in z$

44. **If** $F:R\to[-2,2]$ **then** $F(x)$ **is**

a)one-one b) onto c) into d)None

45. **Which of the following is correct**

a) periods of f(x) ,g(x) and F(x) makes A.P with the common difference $\frac{\pi}{3}$

b) perod of f(x),g(x) and F(x) are same and is equal to 2 π

c)sum of the periods of f(x),g(x) and F(x) is 3π

d) sum of the periods of f(x),g(x) and F(x) is 6π

46. **Which of the following is correct**

a) the domain of G(x) and H(x) are same
b)the range of G(x) and H(x) are same
c) the union of the dmain of G(x) and H(x) are all real numbers
d)None

47. **If the solutions of** $F(x)-G(x)=0$ **are** $x_1,x_2,x_3-----x_n$ **where** $x\in[0,5\pi]$ **then**

a) $x_1,x_2,x_3-----x_n$ are in A.P with common difference $\frac{\pi}{4}$

b) the no.of solutions of

$F(x)-G(x)=0\ is\ 10\forall x\in[0,5\pi]$

c)the sum of all solutions of

$F(x)-G(x)=0\ is\ \ \forall x\in[0,5\pi] is\ 25\pi$

d) b,c are true

(P) Let $f:N\to N$ be a function defined by f(x) = the biggest +ve integer obtained by reshuffling the digits of 'x'. For example f(296)=962. Now answer the following.

48. f is

a) one-one ,onto b) one-one and into
c)many-one and onto d)many-one and into

49. The biggest +ve integer which divides $f(n)-n, \forall n\notin N$ is

a)3 b)9 c)18 d)27

50. The range of f is

a)N
b)set of +ve integers whose digits are non-increasing from left to right
c)set of +ve whoe digits are non-decreaseing from left to right
d)None

(P) A function 'f' from a set X to Y is called onto if every $y\in Y, \exists x\in X$ such that f(x)=y. Unless the contrary is specified, a real function is onto if it takes all real values, Otherwise, it is called on into function. Thus, if X and Y are finite sets , then 'f' can not be onto if Y contains more elements than 'x'. Now answer teh following.

51. The polynomial function

$f(x)=a_0x^n+a_1x^{n-1}+a_2x^{n-2}+.......+a_n$

where $a_0\neq 0$ is onto , for

a) all positive integers n
b) all even +ve integers 'n'
c)all odd +ve integers'n'
d)no +ve interger

52. The function $f(x)=\dfrac{x^2+2x+c}{x^2+4x+3c}$ is onto , if

a) $0<c<2$ b) $0<c<4$

c) $\dfrac{-1}{2}<c<\dfrac{1}{2}$ d) $0<c<1$

53. Which of the following is not true

a)A one-one function from the set $\{a,b,c\}\ to\ \{\alpha,\beta,\gamma\}$ is onto also
b)An onto function from an infinite set to a finite set can not be one-one
c)An onto function is alwasys invertible
d)the functions tanx and cotx are onto

(P) Let $f:R\to R$ be a continuous function such that $f(x)-2f\left(\dfrac{x}{2}\right)+f\left(\dfrac{x}{4}\right)=x^2$.

Now answer the following

54. $f(3)=$

a) f(0) b) 4+f(0) c) 9+f(0) d) 16+f(0)

55. The equation $f(x)-x-f(0)=0$ have exactly

a) no solution b) one solution
c)two solution d)infinite solutions

56. $f^1(0)=$

a)0 b)1 c) $f(0)$ d) $-f(0)$

(P) If $(f(x))^2 f\left(\dfrac{1-x}{1+x}\right)=64x,\ x\neq 0$ then

57. $f(x)=$

a) $4x^{\frac{2}{3}}\left(\dfrac{1+x}{1-x}\right)^{\frac{1}{3}}$ b) $x^{\frac{1}{3}}\left(\dfrac{1-x}{1+x}\right)^{\frac{1}{3}}$

c) $x^{\frac{2}{3}}\left(\dfrac{1-x}{1+x}\right)^{\frac{1}{3}}$ d) $16x^{1/3}\left(\dfrac{1+x}{1-x}\right)^{1/3}$

58. The domain of $f(x)$ is

a) $[0,\infty)$ b) $R-\{-1,1\}$

c) $(-\infty,\infty)$ d) $R-\{0,1,-1\}$

59. The value of $f\left(\dfrac{9}{7}\right)$ is

a) $8\left(\dfrac{7}{9}\right)^{2/3}$ b) $4\left(\dfrac{9}{7}\right)^{1/3}$

c) $-8\left(\dfrac{9}{7}\right)^{2/3}$ d) $-4\left(\dfrac{9}{7}\right)^{1/3}$

(P) Based upon each paragraph, three multiple choice questions have to be answered. Each question has four choices a,b,c and d, out of which only one is correct.

Passage: Consider the functions

$$f(x)=\begin{cases} x+1, & x\le 1 \\ 2x+1, & 1<x\le 2 \end{cases}$$ **and**

$$g(x)=\begin{cases} x^2, & -1\le x<2 \\ x+2, & 2\le x\le 3 \end{cases}$$

60. The domain of the function $f(g(x))$ is

a) $\left[0,\sqrt{2}\right]$ b) $[-1,2]$

c) $\left[-1,\sqrt{2}\right]$ d) None of these

61. The range of the function $f(g(x))$ is

a) $[1,5]$ b) $[2,3]$

c) $[1,2]\cup(3,5]$ d) none of these

62. The number of roots of the equation $f(g(x))=2$ is

a) 1 b) 2

c) 4 d) None of these

(P) Let $f(x)=\sin x - x\cos x, \forall x\in R$. Now answer the following.

63. The least +ve value of 'x' for which f(x)=0 lies in the quadrant

a) Q_1 B) Q_2 C) Q_3 D) Q_4

64. The set of all the values of $x\in(0,2\pi)$ for which $f(x)>0$ is

a) $(0,\pi)$ b) $(\pi,2\pi)$ c) $\left(\frac{\pi}{2},\frac{3\pi}{2}\right)$ d) None

65. If α is the least +ve value for which $\tan\alpha=\alpha$ then the area bounded by $y=f(x)$, $X-axis, x=0$ and $x=2\pi$ is

a) 4 b) $4(1-\cos\alpha)$

c) $4(1+\cos\alpha)$ d) $4-2\left(2+\alpha^2\right)\cos\alpha$

MATRIX-MATCHING QUESTIONS

66. Match the following:

COLUMN-I

a) $f:R\to\left[\frac{\pi}{4},\pi\right)$ and $f(x)=\cot^{-1}\left(2x-x^2-2\right)$ then f(x) is

b) $f:R\to R$ and $f(x)=e^{ax}\sin(bx)$ where $a,b\in R^+$ then f(x) is

c) $f:\to R^+\to[2,\infty)$ and $f(x)=2+3x^2$ then f(x) is

d) $f:X\to X$ and $f(f(x))=x\forall x\in X$ then f(x) is

COLUMN-II

p) one-one

q) into

r) many-one

s) onto

67. Match the following:

LIST-I

a) the integral value of $x\in(-\pi,\pi)$ satisfying the equation $|x^2-1+\cos x|=|x^2-1|+|\cos x|$ can be

b) the number of solutions of $[x]^2=x+2\{x\}$ is (where [.] is G.I.F, {.} is fractional part functions)

c) If $f(x)=\sin^{-1}x+\cos^{-1}x+\tan^{-1}x$ then $[f(x)]$ is

d) An allowable value of $f(x)=\sqrt{\ln(\cos(\sin x))}$

LIST-II

p) 0

q) 1

r) 2

s) 4

t) -1

68. **Match the following equations with their number of roots**

LIST-I | **LIST-II**

a) $x^2 \tan x = 1; x \in [0, 2\pi]$ — p) 5

b) $2^{\cos x} = |\sin x|; x \in [0, 2\pi]$ — q) 2

c) If $f(x)$ is a polynomial of degree 5 with real coefficients such that $f(|x|) = 0$ has 8 real roots then the number of roots of $f(x) = 0$ — r) 3

d) $7^{|x|}(|5 - |x||) = 1$ — s) 4

69. **Match the following:**

COLUMN-I

a) If $f(x) = x + 1$, when $-1 \le x < 0$

$= x^2 - 1; when x \ge 0$ then $(fof)(x)$ for $-1 \le x < 0$ is

b) If $f\left(\frac{2\tan x}{1+\tan^2 x}\right)$

$= \frac{(\cos(2x)+1)(\sec^2 x + 2\tan x)}{2}$ then f(x) is

c) If $f(x+y+1) = \left(\sqrt{f(x)} + \sqrt{f(y)}\right)^2 \forall x,$

$y \in R$ and $f(0) = 1$ then $f(x)$ is

d) If $4 < x < 5$ and $f(x) = \left[\frac{x}{4}\right] + 2x + 2$

(Where [.] is G..I.F) then $f^{-1}(x)$ *is*

COLUMN-II

p) $\frac{x-3}{2}$

q) $x^2 + 2x$

r) 1+x

s) $(x+1)^2$

ASSERTION - REASON QUESTIONS

The following questions consist of two statements, one labelled as 'Statement-I' and the other 'Statement-II'. You are to examine these two statements carefully and decide if the Statement-I and the Statement-II are individually true and if so, whether the Statement-II is the correct explanation for the given Statement-I. Select your answer to these items using the codes given below and then select the correct option.

(A) Both S-I and S-II are individually true and R is the correct explanation of A

(B) Both S-I and S-II are individually true but S-II is not the correct explanation of S-I

(C) S-I is true but S-II is false

(D) S-I is false but S-II is true

70. **Consider two functions** $f(x) = 1 + e^{\cot^2 x}$ **and**

$$g(x) = \sqrt{2|\sin x| - 1} + \frac{1-\cos(2x)}{1+\sin^4 x}$$

S-I: The solution of f(x)=g(x) is given by

$x = (2n+1)\frac{\pi}{2}; \forall n \in I$

S-II: If $f(x) \ge k$ **and** $g(x) \le k$ $(\, where\ K \in R)$ **then the solutions of the equation** $f(x) = g(x)$ **is the solution corresponding to** $f(x) = K$

71. **S-I:** $f: R \to [-1,1], f(x) = \cos x$ **is not a bijection**

S-II: Every even function is not one-one

72. **S-I: The function** $f(x) = |x|$ **is not one-one**

S-II:The negative real numbers are not the pre images of any real numbers

73. **S-I: If** $f(x) = x^5 - 16x + 2$ **then f(x) =0 has only one root in [-1,1]**

S-II:f(-1) and f(1) are of opposite signs

INTEGER QUESTIONS

74. The period of the function satisfying the relation $f(x)+f(x+3)=0,\ \forall x\in R$ is

75. If the period of $\dfrac{\cos(\sin(nx))}{\tan\left(\frac{x}{n}\right)}; n\in N$ is 6π then 'n' is

76. Let $f(x)=x(2-x); 0\le x\le 2$. If the definition of 'f' is extended over the set R-[0,2] by $f(x+1)=f(x)$ then f is a periodic function with period.

77. If $f(x)$ and $g(x)$ are any two real valued functions such that $|f(x)+g(x)| \ge |f(x)|+|g(x)|$ and $g(x)\ne 0$, $f(x)g(x)\le 0$ then the value of $\sum_{r=1}^{100} f(r)=$

78. The period of $f(x)=\sin(\sin(\pi x))+e^{\{3x\}}$ is (where {.} is fractional part function).

79. If $f(x)=\operatorname{sgn}(x^2-2x+3)$ then the value of $f(x)$ is

80. Given $y=2[x]+3$ and $y=3[x-2]+5$ (where [.] dentoes the greatest integer function) then the value of $[x+3y]$ is 37λ then λ is

81. If the range of the function
$$f(x)=\frac{(1+x+x^2)(1+x^4)}{x^3}$$ when $x>0$ is $[K,\infty)$ then K is

82. If fractional parts of $\frac{1}{x}$ and x^2 for some $x\in(\sqrt{2},\sqrt{3})$ are equal then the value of $x^4-\frac{3}{x}$ is

83. A non-zero function $f(x)$ is symmetrical about the line $y=x$ then the value of λ (constant) such that
$$f^2(x)=\left(f^{-1}(x)\right)^2-\lambda x f(x)f^{-1}(x)+3x^2 f(x)\ \forall x\in R^+$$ is

84. If $x^2+4y^2=4$ and range of $f(x,y)$ is $[L,M]$ where $f(x,y)=x^2+y^2-xy$ then LM is

85. If the periodic function $f(x)$ satisfies the equation $f(x+1)+f(x-1)=\sqrt{3}f(x)$ $\forall x\in R$ and the period of $f(x)$ is 4λ then $\lambda=$

KEY - LEVEL-VI

SINGLE ANSWER QUESTIONS

1) B	2) A	3) C	4) B
5) A	6) B	7) C	8) A
9) B	10) C	11) B	12) C
13) C	14) C	15) D	16) C
17) B	18) A	19) C	20) B
21) B	22) A	23) C	24) B
25) A	26) B	27) B	28) C
29) D	30) C	31) B	32) C

MULTIPLE ANSWER QUESTIONS

33)A,B	34)A,B	35) C,D	36) B,C
37) A,B	38) B,D	39)A,B	40)B,C,D
41)A	42)A,C,D		

PARAGRAPH QUESTIONS

43)C	44)B	45)C	46)C
47)D	48)D	49)B	50)B
51)C	52)D	53)C	54)D
55)C	56)A	57)A	58)B
59)C	60)C	61)C	62)B
63)C	64)D	65)D	

MATRIX-MATCHING QUESTIONS

66) A-Q,R; B-R,S; C-P,S; D-P,S
67) A-Q,T; B-S; C-P,Q,R; D-P
68) A-Q; B-S; C-P; D-S
69) A-Q; B-R; C-S; D-P

ASSERTION - REASON QUESTIONS

70)C	**71)A**	**72)C**	**73)B**

INTEGER QUESTIONS

74)6	**75)6**	**76)2**	**77)0**
78)2	**79) 1**	**80) 1**	**81)6**
82)5	**83)3**	**84)3**	**85)3**

HINTS - LEVEL-VI

SINGLE ANSWER QUESTIONS

1. $e^{x^2}+e^{-x^2}>0$ and

$e^{x^2}-e^{-x^2}=2\left\{x^2+\dfrac{x^6}{3}+......\right\}>0$

f(x)>0

$\Rightarrow$ range $\neq$ co-domain $\Rightarrow$ into function

Also

$f^1(x)=\dfrac{8x}{\left(e^{x^2}+e^{-x^2}\right)^2}\Rightarrow f^1(x)>0$ if $x>0$ and $f^1(x)<0$ if $x<0$

Not one-one ----> many-one

2. Even function is symmetrical about the y-axis i.e , x=0

$\Rightarrow f(x)=f(-x)\Rightarrow f(0+x)=f(0-x)$

symmetrical about $x=0$

Similarly symmetrical about

x=a $\Rightarrow f(a+x)=f(a-x)$

3. Put x=a-x on both sides

$\Rightarrow f(x+a)=f(a-x)$

4. $f(-1)=2$

$f(1)=0$ incase of decreasing function

$-a+b=2$

$a+b=0\Rightarrow b=1, a=-1$

$\Rightarrow f(x)=-x+1$

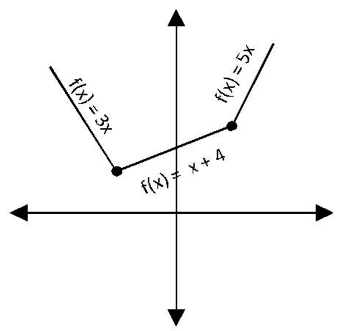
5. Clearly $f^1(x)=e^{3x^2-3x+2}.3\left(x^2-1\right)\geq 0\Rightarrow 'f'$ is increasing

f(x) is one-one

range of f *is* $(f(-\infty), f(-1)]=$

$=\left(0, e^4\right)\neq\left(0, e^5\right]\Rightarrow 'f'$ is into

6.



Let $f(x)=x+|2(x+1)|+2|x-1|$

Then

$$f(x)=\begin{cases}x-2(x+1)-2(x-1), & x<-1\\ x-2(x+1)-2(x-1), & x\leq x\leq -1\\ x-2(x+1)+2(x-1), & x>1\end{cases}$$

$$f(x)=\begin{cases}-3x, & x<-1\\ x+4, & -1\leq x\leq -1\\ 5x, & x>1\end{cases}$$

graph of $y=f(x)$ is as shown. Clearly y=k can intersect $y=f(x)$ at exactly one point only if k=3.

7. $f(2)=2\sum_{r=1}^{1}f(1)=2$

$f(3)=2\sum_{r=1}^{2}f(r)=2[f(1)+f(2)]=2[1+2]=6$

$f(4)=18\rightarrow\sum_{n=1}^{m}f(n)=3^{m-1}$

8. $1-\log_{10}\left(x^2-5x+6\right)>0$ and

$x^2-5x+16>0$

$\log_{10}\left(x^2-5x+6\right)<1$

$x^2-5x+16<10$

$x \in (2,3)$

this is true for all $x \in R$ $(\because Disc < 0)$

$and\ leading\ coeff > 0$

9. $-1 \le x+[x] \le 1$

when $x \in Z$ i.e Let x=k

$\Rightarrow -1 \le 2k \le 1 \Rightarrow \frac{-1}{2} \le k \le \frac{1}{2} \Rightarrow k = 0 \Rightarrow x=0$

when $x \notin Z$ i.e let x=k+f $\Rightarrow -1 \le 2k+f \le 1$

$\Rightarrow \frac{-1-f}{2} \le k \le \frac{1-f}{2} \Rightarrow k = 0 \Rightarrow 0 < x < 1.$

$\therefore Df = [0,1)$

10. $x^2 - [x^2] > 0$

$x^2 > [x^2]$

$x^2 \in R\ and\ x^2 \ne 0,1,2-----$

$\therefore Domain\ is\ R - \{\pm\sqrt{n}, n \in z\ and\ n \ge 0\}$

11. $[\cos^{-1} x] \ne 1\ and [\cos^{-1} x] > 0$

$\cos^{-1} x \notin [1,2)$ $\qquad [\cos^{-1} x] = 2$

$2 \le \cos^{-1} x \le \pi, \cos n \ge x \ge -1, -1 \le x \le \cos 2$

12. Take $-1 \le x - x^2 \le 1$ and $1 - \frac{1}{|x|} \ge 0$

$and\ x^2 - 1 < 1; x^2 \ge 2$

13. least value of $x^2 + x + 1 = \frac{3}{4}$ at $x = \frac{1}{2}$

$f(-1) = 2$

$f(1) = 0$ in case of decreasing function

$-a + b = 2, a + b = 0$

$\Rightarrow b = 1,\ a = -1$

$\Rightarrow f(x) = -x + 1$

$\therefore$ least value of $\sqrt{x^2 + x + 1}$ is $\frac{\sqrt{3}}{2}$

Also $\sqrt{x^2 + x + 1} \le 1$

Thus $\frac{\sqrt{3}}{2} \le \sqrt{x^2 + x + 1} \le 1$

$\Rightarrow \sin^{-1}(\frac{\sqrt{3}}{2}) \le \sin^{-1}(\sqrt{x^2 + x + 1}) \le \sin^{-1}(1)$

$\Rightarrow \frac{\pi}{3} \le \sin^{-1}(\sqrt{x^2 + x + 1}) \le \frac{\pi}{2}$

14. $f(x) = -px + q - rx; x \le 0$

$= f(x) = -px + q + rx; 0 < x \le \frac{q}{p}$

$= f(x) = px - q + rx; \frac{q}{p} < x$

$for\ r = p;\ f^1(x) < 0; x \le 0$

$= 0, if\ 0 < x < \frac{q}{p},\ > 0\ if\ x > \frac{q}{p}$

from the graph(i) we have infinitely many points for min.value of f(x)

$for\ r < p\quad f^1(x) < 0\ if\ x \le 0$

$< 0\ if\ 0 < x \le \frac{q}{p}$

$> 0\ if\ x > \frac{q}{p}$

from the graph(ii), only one point of minimum x=q/p

$for\ r > p\quad f^1(x) < 0\ ;\ if\ x \le 0$

$> 0;\ if\ x > 0$

from the graph(iii), onley one point of minimum

15. $f(x) = e^{x^2} + e^{-x^2} \to f^1(x)$

$= 2x\left(e^{x^2} + e^{-x^2}\right) \ge 0 \forall x [0,1]$

f(x) is a non-decreasing function on $[0,1]$

Hence max.value of $f(x) = f(1) = e + \frac{1}{e} = a$

$g^1(x) = (2x^2 + 1)e^{x^2} - 2x.e^{-x^2}$

$\ge 0 \forall x \in [0,1] \to g(x)$

is an increasing function [0,1]

16. Let $y = f(x) = \cos^{-1}\left(\frac{\sqrt{2x^2 + 1}}{x^2 + 1}\right)$

Clearly $y \in (0, \pi)$ -----(1)

$Also\ \cos y = \frac{\sqrt{2x^2 + 1}}{x^2} + 1 > 0$

$\cos y > 0$

$y < \frac{\pi}{2}$ -------(2)

from (1) and (2) $y \in \left[0, \frac{\pi}{2}\right) = R_f$

17. f(x)=1 is false; $f(y)\neq 1$ is false and $f(z)\neq 2$ is true then Injection is possible
i.e $f(x)=2(or)3$; $f(y)=1; f(z)=1(or)3$

18. $f(x)=1-x; 0\le x\le 2$
$=x-3; 2<x\le 4$
$g(x)=2+x; -1\le x\le 0$
$=2-x; 0<x\le 3$
$(fog)(x)=-1-x; -2\le x\le 0; -1\le x\le 0$
$=x-1; 0<x\le 2; -1\le x\le 0$
$=x-1; 0\le x\le 2; 0<x\le 3$
$=-x-1; -2\le x<0; 0<x\le 3$

19. Let $y=f(x)\rightarrow y=\dfrac{x^2+1}{x}\rightarrow x^2-xy+1=0$
$\rightarrow x=\dfrac{y\pm\sqrt{y^2-4}}{2}$
$f^{-1}=\dfrac{y\pm\sqrt{y^2-4}}{2}$
$f^{-1}(x)=\dfrac{x\pm\sqrt{x^2-4}}{2}$
$\therefore$ the range of $f^{-1}(x)$ is $[1,\infty)$, ignore the negative sign
$\therefore f^{-1}(x)=\dfrac{x+\sqrt{x^2-4}}{2}$

20. $f(x)+f(x+4)$
$=f(x+2)+f(x+6)\forall x\in R$
Replace x by $x+2$ and simplify
On solving we get $f(x)=f(x+8)$

21. $f(x)+f(x+3)=0----(1)$
replace 'x' by 'x+3'
$f(x+3)+f(x+6)=0\ \ ----(2)$
$(1)-(2)\rightarrow f(x)=f(x+6)$
$period\ of\ f(x)\ is\ 6$
$g^1(x)=f(x+n)=f(x+(n-3)+3)$
$=f(x+(n-3))(\because period\ of\ 'f'\ is\ 3)$
$=f(x+3)=f(x)\rightarrow g^1(x)=f(x)$
$\rightarrow g^1\left(\dfrac{7}{3}\right)=f\left(\dfrac{7}{3}\right)=f\left(\dfrac{-2}{3}+3\right)=f\left(\dfrac{-2}{3}\right)=7$

22. $x^2f(x)-2.f\left(\dfrac{1}{x}\right)=g(x)\Rightarrow$ replacing 'x' by $\dfrac{1}{x}$, then $\dfrac{1}{x^2}.f\left(\dfrac{1}{x}\right)-2f(x)=g\left(\dfrac{1}{x}\right)$ (or)
$2f\left(\dfrac{1}{x}\right)-4x^2f(x)=2x^2.g\left(\dfrac{1}{x}\right)$ --------(2)
$(1)+(2)\Rightarrow\ \ f(x)=-\left(\dfrac{g(x)+2x^2.g\left(\dfrac{1}{x}\right)}{3x^2}\right)$
$f(-x)=-\left(\dfrac{g(-x)+2x^2.g\left(-\dfrac{1}{x}\right)}{3x^2}\right)$
$f(x)=-f(x)$
($\therefore g(x)$ is an odd and f(x) is an even function)
$f(x)=0\Rightarrow f(5)=0$

23. $2f(xy)=(f(x))^y+(f(y))^x$ replacing 'y' by '1', then we get
$2f(x)=f(x)+(f(1))^x$
$\Rightarrow f(x)=a^x$
Now $(a-1)\sum_{i=1}^{n}f(i)=(a-1)$
$\sum_{i=1}^{n}a^i=(a-1)\left[a+a^2+a^3+------+a^n\right]$
$=(a-1)\left(\dfrac{a(a^n-1)}{a-1}\right)=a^{n+1}-a$

24. $f(xf(y))=x^py^q\Rightarrow\left[f(xf(y))\right]^{\frac{1}{p}}=x.y^{\frac{q}{p}}$
$x=\dfrac{\left[f(xf(y))\right]^{1/p}}{y^{q/p}}---(1)$
Now let $x.f(y)=1$
$x=\dfrac{1}{f(y)}.....(2)$
Hence $f(x.y^{q/p})=x^p.y^q$
put $y^{q/p}=z$
$f(xz)=(xz)^p$

$\Rightarrow f(\lambda)=\lambda^p ----(4)$

$from\ (3)\ and\ (4)\ y^p=y^{q/p}$

$\Rightarrow p=\frac{q}{p}\Rightarrow p^2=q$

$f(y)=\frac{y^{q/p}}{(f(1))^{1/p}}\Rightarrow f(1)=\frac{1}{(f(1))^{1/p}}$

$\Rightarrow f(1)=1$

then $f(y)=y^{q/p} ----(3)$

25. $f(x+y)+f(x-y)$

$=2f(x)f(y)------(1)$

replace 'x' by 'y' and 'y' by 'x' in ---(1)

$f(y+x)+f(y-x)$

$=2f(y)f(x)----(2)$

from (1) and (2) , we get

$f(y-x)=f(x-y)$ putting $y=2x$ then

$f(x)=f(-x)$

Hence $f(x)$ is an even function.

26. Replace 'x' by 'a' and 'y' by 'x-a'

$\Rightarrow f(a-(x-a))=f(a).f(x-a)-f(0).f(x)$ $----(1)$

put $x=0;y=0\rightarrow f(0)=[f(0)]^2-[f(a)]^2$

$\Rightarrow f(a)=0,\ [\because f(0)=1]$

$from\ (1), f(2a-x)=-f(x)$

27. $3f(x)+2f\left(\frac{x+59}{x-1}\right)=10x+30$

$put\ x=7$

$3f(7)+2f(11)=70+30=100$

$put\ x=11$

$3f(11)+2f(7)=140\rightarrow f(7)=4$

28. $f(x+a)=\frac{1}{2}+\sqrt{f(x)-(f(x))^2}$,

$put\ x=x+a$

$f(x+a+a)=\frac{1}{2}+\sqrt{f(x+a)-(f(x+a))^2}$

$f(x+2a)=\frac{1}{2}+\sqrt{f(x+a)\{1-f(x+a)\}}$

$=\frac{1}{2}+\sqrt{\left\{\frac{1}{2}+\sqrt{f(x)-(f(x))^2}\right\}\left\{\frac{1}{2}-\sqrt{f(x)-(f(x))^2}\right\}}$

$=\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2+f(x)-(f(x))^2}$

$=\frac{1}{2}+f(x)-\frac{1}{2}$

$=f(x)\qquad \because f(x)\geq 2$

$f(x+2a)=f(x)$

$period\ of\ 'f' is\ 2a$

29. $f(x,y)=f(2x+2y,2y-2x)\forall x,$

$y\in R------(1)$

$f(2(2x+2y)+2(2y-2x),2(2y-2x)-2(2x+2y))$

$f(x,y)=[from(1)]$

$f(x,y)=f(8y-,8x)----(2)$

$=f(8(-8x),-8(8y))\ [from\ (2)]$

$f(x,y)=f(-64x,-64y)---(3)$

$=f(-64(-64x),-64(-64y))\ from(3)$

$f(x,y)=f(2^{12}x,2^{12}y)$

replace 'y' by 0 $f(x,0)=f(2^{12}x,0)$

replace 'x' by 2^y

$f(2^y,0)=f(2^{12}.2^y,0)$

$=f(2^{12+y},0)$

$f(2^x,0)=f(2^{12+x},o)$

$g(x)=g(12+x)\ for\ all\ 'x'$

$period\ of\ 'g'\ is\ 12$

30. $f(2)=\min(4,2,\operatorname{sgn}(17))=\min(4,2,1)=1$

31. $f(x)=\frac{5}{4}$ *on simplification*

Now $g(f(x))=g\left(\frac{5}{4}\right)=1\Rightarrow y=1$ is a straight line

32. In the sum 1 occurs n times, $\frac{1}{2}$ occurs (n-1) times , $\frac{1}{3}\ occurs(n-2)$ times and so on

$\therefore f_1(1)+f_2(2)+f_3(3)+........+f_1(n)$

$=n(1)+(n-1)\frac{1}{2}+(n-2)\frac{1}{3}+----+1.\frac{1}{n}$

$=n\left(1+\frac{1}{2}+\frac{1}{3}+------+\frac{1}{n}\right)$

$-\left(\frac{1}{2}+\frac{2}{3}+\frac{3}{4}+-----+\frac{n-1}{n}\right)$

$=n.f_1(n)-\left[\left(1-\frac{1}{2}\right)+\left(1-\frac{1}{3}\right)+\left(1-\frac{1}{4}\right)+----+\left(1-\frac{1}{n}\right)\right]$

$=n.f_1(n)-[n-f_1(n)]=(n+1)f_1(n)-n$

MULTIPLE ANSWER QUESTIONS

33. Even extension is f(-x)
Odd extension is -f(-x)

34. $(fog)(x)=\sqrt{3|\sin x|-\sin x-2}$

$\Rightarrow 3|\sin x|-\sin x-2\geq 0$

If $\sin x>0\Rightarrow \sin x\geq 1$

$\sin x=1\Rightarrow x\in 2n\pi+\frac{\pi}{2}$

$\sin x<0\Rightarrow -1\leq \sin x\leq \frac{1}{2}$

$x\in\left[2n\pi+\frac{7\pi}{6},2n\pi+\frac{11\pi}{6}\right]$

35. $|[|x|-1]|-5>0$

$\Rightarrow |x|-1\geq 6\ (or)\ |x|-1<-5$

$\Rightarrow x\in(-\infty,-7]\cup[7,\infty)$

36. $2f(\sin x)+f(\cos x)=x$ ----(1)

replace 'x' by $'\frac{\pi}{2}-x'$,we get

$2f(\cos x)+f(\sin x)=\frac{\pi}{2}-x----(2)$

from (1) *and* (2)

$f(\sin x)=x-\frac{\pi}{6}$

let $\sin x=y\Rightarrow x=\sin^{-1}y$

$f(y)=\sin^{-1}y-\pi/6$

Domain $=[-1,1]$

$-\frac{\pi}{2}\leq \sin^{-1}x\leq\frac{\pi}{2}$

$-\frac{2\pi}{3}\leq \sin^{-1}x-\frac{\pi}{6}\leq\frac{\pi}{3}$

37. $(fog)(x)=\sin(\ln|x|)\Rightarrow R_1=[-1,1]$

$(gof)(x)=\ln|\sin x|\Rightarrow R_2=(-\infty,0]$

38. Let $q(x)=K$ *where* $K\in R$ and

$p(x)=a_0+a_1x+a_2x^2+a_3x^3+------a_nx^n=k$

$a_0=a_2=a_3------a_n=0$ *and* $a_1=1$

p(x)--->odd and of odd degree

39. $f(x)=1+\sin x;0\leq x\leq\frac{3\pi}{4}$

$=1-\cos x;\frac{3\pi}{4}\leq x\leq\frac{3\pi}{2}$

$=1;\frac{3\pi}{2}\leq x\leq 2\pi$

$g(x)=1-x;x\leq 0$

$=1;0\leq x\leq 2$

$=x-1;x\geq 2$

$f(0)=1\Rightarrow g(f(1))=1$

$f(1)=1+\sin 1$

and $f(g(0))=1+\sin 1$

40. *If* $f(x)$ *is periodic*

if $4-a^2$

is a perfect square i.e $\sqrt{4-a^2}$ is a rational number.

$\Rightarrow a=0,2,-2$

41. $f(x)=f^1(x).f^{11}(x)$

degree of $f(x)$ *is* 3

$n=n-1+n-2-->n=3$

f(x) is an onto function

42. Clearly

$f(1+x)=f(1-x)--->$ symmetric about the line $x=1$

max.value of $\ln(2x-x^2)$ as well as that of

$\sin\left(\frac{\pi x}{2}\right)$ ocurs at x=1

$Domain \Rightarrow 2x - x^2 > 0 \Rightarrow x^2 - 2x < 0$

$x \in (0,2)$ as $x \to 0+ (or) x \to 2-$

$f(x) \to -\infty$

PARAGRAPH QUESTIONS

43. H(x)= $\frac{f(x)}{g(x)} = \frac{1-2\sin^2 x}{\cos 2x} = 1\ but\ \cos(2x) \neq 0$

$2x \neq n\pi + \frac{\pi}{2}$

$Domain = R - \{(2n+1)\frac{\pi}{4}\}$

$and Range = 1$

44. F(x)=f(x)+g(x)=

$1 - 2\sin^2 x + \cos 2x = 2\cos(2x)$

$-1 \le \cos(2x) \le 1 \Rightarrow -2 \le 2\cos(2x) \le 2$

$\Rightarrow onto\ function$

45. $f(x) = 1 - 2\sin^2 x$ with period π

$g(x) = \cos(2x)$ with period π

$F(x) = 2\cos 2x$ with period $\pi \Rightarrow \pi + \pi + \pi = 3\pi$

46. G(x)=f(x)-g(x)=

$1 - 2\sin^2 x - \cos(2x) = 0$

$H(x) = 1,$ $union\ \ of\ their\ \ Domains\ is\ R$

47. $F(x) - G(x) \to 2g(x) = 0$

$\to \cos(2x) = 0 \to x$

$= \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, ------, \frac{19\pi}{4}$

No.of solutions=10 and their sum $= 25\pi$

48. As f(296) =f(926)---> many-one
Further, the number whose digits increase from left to right
eg: 159 have no pre-image---> into

49. Here f(n) and n leave the same remainder, when divided by 9. The remainder, when a +ve integer is divided by 9 is same as the sum of the digits of the number (until the sum becomes a single digit number

50. Clear from the definition

51.If the degree is odd, then $f(-\infty) = -\infty; f(\infty) = \infty$

52. Let $y = \frac{x^2 + 2x + c}{x^2 + 4x + 3c}$

$\Rightarrow (y-1)x^2 + (4y-2)x + (3(y-2)) = 0.$

Here $\Delta \ge 0$

$(4-3c)y^2 - 4 - (1-c)y + (1-c) \ge 0$

$4 - 3c > 0$ and

$16(1-c)^2 - 4(4-3c)(1-c) \le 0$

$0 < c < 1$

53. conceptual

54. $f(x) - 2f\left(\frac{x}{2}\right) + f\left(\frac{x}{4}\right) = x^2$

Replace x by $\frac{x}{2}$

$f\left(\frac{x}{2}\right) - 2f\left(\frac{x}{4}\right) + f\left(\frac{x}{8}\right) = \left(\frac{x}{2}\right)^2$

$f\left(\frac{x}{4}\right) - 2f\left(\frac{x}{8}\right) + f\left(\frac{x}{16}\right) = \left(\frac{x}{4}\right)^2$

$f\left(\frac{x}{2^n}\right) - 2f\left(\frac{x}{2^{n+1}}\right) + f\left(\frac{x}{2^{n+2}}\right) = \left(\frac{x}{2^n}\right)^2$

Adding we get

$f(x) - f\left(\frac{x}{2}\right) - f\left(\frac{x}{2^{n+1}}\right) + f\left(\frac{x}{2^{n+2}}\right)$

$= x^2\left(1 + \frac{1}{2^2} + ------- + \frac{1}{2^{2n}}\right)$

As $n \to \infty$

$f(x) - f\left(\frac{x}{2}\right) = \frac{4x^2}{3}$

repeating we get

$f(x) - f(0) = \frac{16x^2}{9} \Rightarrow f(3) = 16 + f(0)$

55. $f(x) - f(0) - x = 0$

$\Rightarrow \frac{16x^2}{9} - x = 0 \Rightarrow x = 0, \frac{9}{16}$

56. $f^1(x) = \frac{32x}{9} \Rightarrow f^1(0) = 0$

57,58,59.

$(f(x))^2 . f\left(\frac{1-x}{1+x}\right) = 64x ----(1)$

put $\frac{1-x}{1+x} = y$ and $x = \frac{1-y}{1+y}$ $we\ get$

$\left\{f\left(\frac{1-y}{1+y}\right)\right\}^2 f(y) = 64\left(\frac{1-y}{1+y}\right)$

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$$\Rightarrow f(x).\left\{f\left(\frac{1-x}{1+x}\right)\right\}^2 = 64\left(\frac{1-x}{1+x}\right)----(2)$$

from $(1)^2$ *and* (2), *we have*

$$\frac{(f(x))^4\left\{f\left(\frac{1-x}{1+x}\right)\right\}^2}{f(x)\left\{f\left(\frac{1-x}{1+x}\right)\right\}^2} = \frac{(64x)^2}{64\left(\frac{1-x}{1+x}\right)}$$

$$\Rightarrow (f(x))^3 = 64x^2\left(\frac{1+x}{1-x}\right)$$

$$\Rightarrow f(x) = 4.x^{2/3}\left(\frac{1+x}{1-x}\right)^{1/3}$$

Now solve other two questions

60-62

$$f(x) = \begin{cases} x+1, & x \le 1 \\ 2x+1, & 1 < x \le 2 \end{cases}$$

$$g(x) = \begin{cases} x^2, & 1 \le x < 2 \\ x+2, & 2 \le x < 3 \end{cases}$$

$$\Rightarrow \quad f(g(x)) = \begin{cases} g(x)+1, & g(x) \le 1 \\ 2g(x)+1, & 1 < g(x) \le 2 \end{cases}$$

$$\Rightarrow f(g(x)) = \begin{cases} x^2+1, & x^2 \le 1, & -1 \le x < 2 \\ x+2+1, & x+2 \le 1, & 2 \le x \le 3 \\ 2x^2+1, & 1 < x^2 \le 2, & -1 \le x < 2 \\ 2(x+2)+1, & 1 < x+2 \le 2, & 2 \le x \le 3 \end{cases}$$

$$\Rightarrow f(g(x)) = \begin{cases} x^2+1, & -1 \le x \le 1 \\ 2x^2+1, & 1 < x \le \sqrt{2} \end{cases}$$

60. **Hence, the domain of** $f(x)$ **is** $\left[-1, \sqrt{2}\right]$

61. **For** $-1 \le x \le 1$ **we have**

$x^2 \in [0,1] \Rightarrow x^2+1 \in [1,2]$

For $1 < x \le \sqrt{2}$**, we have**

$x^2 \in (1,2] \Rightarrow 2x^2+1 \in (3,5]$

Hence, the range is $[1,2] \cup (3,5]$

62. **For** $f(g(x)) = 2 \Rightarrow x^2+1 = 2$ **and**

$2x^2+1 = 2 \Rightarrow x = \pm 1$ **or** $x = \pm\frac{1}{\sqrt{2}}$

$\Rightarrow \quad x = \pm 1$ **only. Hence, 2 roots**

63. $f(x) = 0 \Rightarrow \tan x = x$

Draw the graphs of

$y = \tan x$ & $y = x$

64. *for* $x \in \left(0, \frac{\pi}{2}\right), f(x) > 0$

as $\tan x > x$

65. $\operatorname{Re}qd\ Area == \int_0^{\alpha} f(x)dx - \int_{\alpha}^{2\pi} f(x)dx$

$-\left[2\cos x + x\sin x\right]_0^{\alpha} + \left[2\cos x + x\sin x\right]_{\alpha}^{2\pi}$

$= 4 - 2\left(2+\alpha^2\right)\cos\alpha$

MATRIX-MATCHING QUESTIONS

66. $a--->$ Let $y = f(x) = \cot^{-1}\left(2x - x^2 - 2\right)$

clearly $y \in (0, \pi)----(1)$

and

$\cot y = 2x - x^2 - 2 = -1 - (x-1)^2 \le 1$

$\cot y \le -1$

$y \in \left(\frac{3\pi}{4}, \pi\right)--->$ into

more over $f^1(x)$ changes its sign in the neighbourhood of x=2

many-one

$b---> \qquad -e^{ax} \le e^{ax}\sin(bx) \le e^{ax} -->$

$many-one, onto$

$c---> f^1(x) = x---> f^1(x) > 0--->$

$many-one\ i.e, f(x)$ is an increasing function

$range = (f(0), f(\infty))$

$= (2, \infty) \ne co-domain---> $ into

d---> clearly identity function--->one-one and onto

67. $a \rightarrow \left(x^2-1\right)\cos x \ge 0 \rightarrow x^2 - 1 \ge 0$ and

$\cos x \ge 0$ (or) $x^2 - 1 \le 0$ and $\cos x \le 0$

$-\frac{\pi}{2} \le x \le -1\ (OR)\ 1 \le x \le \frac{\pi}{2}$

$b \rightarrow [x]^2 = x + 2\{x\} \rightarrow \{x\} = \frac{[x]^2 - [x]}{3}$

Now $0 \le \frac{[x]^2 - [x]}{3} < 1$

$\rightarrow [x] \in \left(b \rightarrow [x]^2 = x + 2\{x\} \rightarrow \{x\} = \frac{[x]^2 - [x]}{3}\right]$

$\rightarrow [x] = -1, 0, 1, 2$

$\{x\} = \frac{2}{3}, 0, 0, \frac{2}{3} \rightarrow x = \frac{-1}{3}, 0, 1, \frac{8}{3}$

$c \rightarrow -1 \le x \le 1 \rightarrow -\frac{\pi}{4} \le \tan^{-1} x \le \frac{\pi}{4}$

$\therefore f(x) = \frac{\pi}{2} + \tan^{-1} x \rightarrow \frac{\pi}{4} \le f(x) \le \frac{3\pi}{4}$

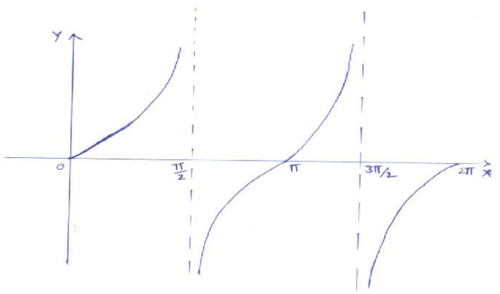
$\rightarrow [f(x)] = 0, 1, or\ 2$

$d \rightarrow \ln(\cos(\sin x)) \ge 0$

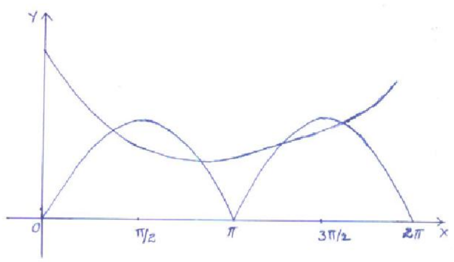
$\rightarrow \cos(\sin x) \ge 1 \rightarrow \cos(\sin x = 1)$

$\rightarrow \ln(\cos(\sin x)) = 0$

68. $a \rightarrow$ draw the graphs of $y = \tan x$ & $y = \frac{1}{x^2}$ from the graph , it is clear that it will have two real roots



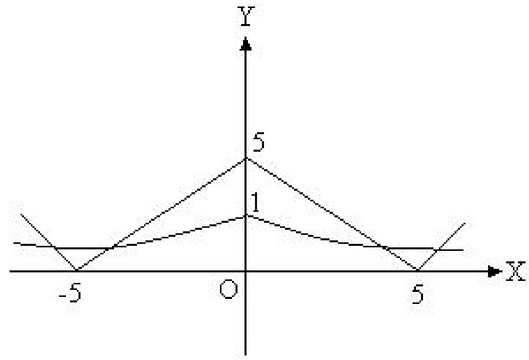
$b \rightarrow$ draw the gaphs $y = 2^{\cos x}$ and $y = |\sin x|$ no.of P.O.I=4



c-->$\because f(|x|) = 0$ has 8 real roots $\rightarrow f(x) = 0$ has 4 real roots $\because f(x)$ is a polynomial of degree 5 , f(x) cannot have even no.of real roots

$f(x)$ has all the five roots real and one roots

$f(x)$ has all the five roots real and one root is negative



$d \rightarrow$ Draw the graphs of $y = 7^{-|x|}$ and $y = |5 - |x||$

69. $a \rightarrow$ when $-1 \le x < 0$ $f(x) \in [0,1)$ and $f(x) = x + 1$

$\therefore f(f(x)) = (x+1)^2 - 1 = x^2 + 2x$

$b \rightarrow \frac{[\cos(2x) + 1][\sec^2 x + 2\tan x]}{2}$

$= \frac{1 + \tan^2 x + 2\tan x}{1 + \tan^2 x}$

$i.e\ f\left(\frac{2\tan x}{1 + \tan^2 x}\right) = 1 + \frac{2\tan x}{1 + \tan^2 x}$

$\therefore f(x) = 1 + x$

$c \rightarrow put\ x = y = 0$

$\rightarrow f(1) = (1+1)^2 = 2^2;$

$put\ x = 0; y = 1 \rightarrow f(2) = 3^2$

$d \rightarrow when\ 4 < x < 5 \rightarrow f(x)$

$= 2x + 3 \rightarrow f^{-1}(x) = \frac{x-3}{2}$

$f(\theta) = \frac{4\sin^2\theta\cos\theta - \cos\theta + \sin\theta}{4(\cos\theta + \sin\theta)}$

$= \frac{1}{2}\sin\theta(\sin\theta + \cos\theta) - \frac{1}{4}$

$= \frac{1}{4}[\sin(2\theta) - \cos(2\theta)] \in \left[\frac{-1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}\right]$

ASSERTION - REASON QUESTIONS

70. $f(x)=g(x)\to L.H.S=1+e^{\cot^2 x}\geq 2$

$As\ \sqrt{2|\sin x|-1}\leq and$

$\frac{1-\cos(2x)}{1+\sin^4 x}=\frac{2\sin^2 x}{1+\sin^4 x}$

$=\frac{2}{\frac{1}{\sin^2 x}+\sin^2 x}\leq 1$

$\therefore RHS\leq 2$ $\therefore$ Equality holds good L.H.S=R.H.S =2

Which is possible when

$\cot^2 x=0$ and $|\sin x|=1$

$\to x=(2n+1)\frac{\pi}{2};n\in I$

$\therefore S_1$ is correct

Clearly $\therefore S_2$ need not be correct always

$\therefore S_1$ *is true and* S_2 *is false*

71. Conceptual
72. Conceptual
73. f(-1) =17 and f(1)=-13

clearly S_1 is also true

INTEGER QUESTIONS

74. $f(x)+f(x+3)=0$ --(1)

replace ‘x’ by ‘x+3’

$f(x+3)+f(x+6)=0------(2)$

$(1)-(2)\Rightarrow f(x)=f(x+6)$

$\Rightarrow period\ of\ f(x)\ is\ 6$

75. Period of $\cos(\sin(nx))$ *is* $\frac{\pi}{n}$ and the period of

$\tan\left(\frac{x}{n}\right)$ *is* πn

Now L.C.M of $\frac{\pi}{n},\pi n$ is $\pi n\ \Rightarrow n=6$

76. $f(x+2)=f(x+1+1)=f(x+1)=f(x)$

Period =2

77. $|f(x)+g(x)|\leq|f(x)|+|g(x)|$ -----(1)

but given that

$|f(x)+f(x)|\geq|f(x)|+|g(x)|$ - - (2)

from (1) and (2)

$|f(x)+g(x)|=|f(x)|+|g(x)|$

this is possible only when

$f(x).g(x)\geq 0$ but given that

$f(x).g(x)\leq 0$

$\to f(x)g(x)=0\to f(x)=0(\because g(x)\neq 0)$

$\therefore \sum_{r=1}^{100} f(r)=0$

78. Conceptual

79: $b^2-4ac<0$.

80. **Properties of G.I.F.**

81. $f(x)=\frac{1+x^4+x+x^5+x^2+x^6}{x^3}$

$=\left(\frac{1}{x^3}+x^3\right)+\left(x+\frac{1}{x}\right)+\left(x^2+\frac{1}{x^2}\right)$

$\geq 2\sqrt{\frac{1}{x^3}.x^3}+2\sqrt{x.\frac{1}{x}}+2\sqrt{x^2.\frac{1}{x^2}}=6$

$\left(use\ a+\frac{1}{a}\geq 2,a>0\right)$

$\Rightarrow f(x)\geq 6$

$\therefore R_f=[6,\infty)$.

82. $\because\quad x\in(\sqrt{2},\sqrt{3})$

$\Rightarrow\ \sqrt{2}<x<\sqrt{3}\quad or\quad \frac{1}{\sqrt{3}}<\frac{1}{x}<\frac{1}{\sqrt{2}}$

$\therefore\ \left\{\frac{1}{x}\right\}=\frac{1}{x}$ **and** $2<x^2<3$

or $0<x^2-2<1$

$\therefore\ \{x^2\}=x^2-2$

Given $\left\{\frac{1}{x}\right\}=\{x^2\}\Rightarrow\frac{1}{x}=x^2-2\qquad ...(i)$

$\Rightarrow x^3-2x-1=0\Rightarrow(x+1)(x^2-x-1)=0$

$x\neq 1$

$\therefore x^2-x-1=0 \Rightarrow x^2=x+1$

$\Rightarrow x^4=x^2+2x+1=x+1+2x+1$

$\Rightarrow x^4=3x+2 \quad ....(ii)$

$\therefore x^4=3x+2$

$\Rightarrow x^4-\frac{3}{x}=(3x+2)-3(x^2-2)$

(from eqs. (i) and (ii))

$=(3x+2)-3(x+1-2)$

=5. $\quad (\because x^2=x+1)$

83. **Given** $f(x)\neq 0$

$\because$ $f(x)$ **is symmetrical about the line** $y=x$**, then**

$$f^{-1}(x)=f(x)=y=x$$

$$\Rightarrow f^2(x)=\left(f^{-1}(x)\right)^2-\lambda x f(x) f^{-1}(x)+3x^2 f(x)$$

$\Rightarrow x^2=x^2-\lambda x.x.x+3x^2.x$

$\Rightarrow \lambda x^3=3x^3$.

$\therefore \quad \lambda=3, x\neq 0$.

84. $\because \quad x^2+4y^2=4$

$$\Rightarrow \left(\frac{x}{2}\right)^2+\left(\frac{y}{1}\right)^2=1.$$

Let $x=2\cos\theta, y=\sin\theta$

$$\Rightarrow f(\theta)=(2\cos\theta)^2+(\sin\theta)^2-(2\cos\theta)(\sin\theta)$$

$=4\cos^2\theta+\sin^2\theta-\sin 2\theta$

$=3\cos^2\theta+(\cos^2\theta+\sin^2\theta)-\sin\theta$

$$=\frac{3}{2}(1+\cos 2\theta)+1-\sin 2\theta=\frac{3}{2}\cos 2\theta-\sin 2\theta+\frac{5}{2}$$

$$\Rightarrow \frac{5}{2}-\sqrt{\left(\frac{3}{2}\right)^2+(-1)^2}\leq \frac{3}{2}\cos 2\theta-\sin\theta$$

$$\frac{5}{2}\leq\frac{5}{2}+\sqrt{\left(\frac{3}{2}\right)^2+(-1)^2}$$

$$\Rightarrow \frac{5-\sqrt{13}}{2}\leq f(\theta)\leq\frac{5+\sqrt{13}}{2}.$$

Hence, $L=\frac{5-\sqrt{13}}{2}$ *and* $M=\frac{5+\sqrt{13}}{2}$.

$$\therefore LM=\left(\frac{5-\sqrt{13}}{2}\right)\left(\frac{5+\sqrt{13}}{2}\right)=\frac{25-13}{4}$$

$$=\frac{12}{4}=3.$$

85: **We have**

$f(x+1)+f(x-1)=\sqrt{3}f(x)\,\forall x\in R$ **...(i)**

Replacing x **by** $x-1$ **and** $x+1$ **in (i),then**

$f(x)+f(x-2)=\sqrt{3}f(x-1)\ ...(ii)$

and $f(x+2)+f(x)=\sqrt{3}f(x+1)\ ...(iii)$

Adding (ii) and (iii), we get

$2f(x)+f(x-2)+f(x+2)$

$=\sqrt{3}\left(f(x-1)+f(x+1)\right)$

$\Rightarrow 2f(x)+f(x-2)+f(x+2)=\sqrt{3}.\sqrt{3}f(x)$

[From (i)]

$\therefore f(x+2)+f(x-2)=f(x) \quad .....(iv)$

Replacing x **by** $x+2$ **in equation (iv),**

then $f(x+4)+f(x)=f(x+2)$ **....(v)**

Adding equations (iv) and (v), we get

$f(x+4)+f(x-2)=0$ **.... (vi)**

Again replacing x **by** $x+6$ **in (vi), then**

$f(x+10)+f(x+4)=0$ **....(vii)**

Subracting (vi) from (vii), we get

$f(x+10)-f(x-2)=0$ **....(viii)**

Replacing x **by** $x+2$ **in (viii), then**

$f(x+12)-f(x)=0$

or $\quad f(x+12)=f(x)$

Hence $f(x)$ **is periodic function with period 12.**